EXTENDING A NON-MINIMAL-PRIME ELEMENT TO A SYSTEM OF PARAMETERS INSIDE A GIVEN POWER OF THE MAXIMAL IDEAL

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Allowed standard prerequisites (editorial).

- If f is not contained in any minimal prime of a Noetherian local ring of dimension d then dim(R/fR) = d - 1. invoked at proof lines 38095-38096. Makes R/fR a (d-1)-dimensional local ring.
- Every Noetherian local ring admits a system of parameters of length equal to its dimension. invoked at proof lines 38097-38099. Provides the parameters of R/fR to lift.

Source reference. Original lemma 00KW: sources/stacks-chow/tags.txt, source lines 766–766. Cited at proof lines 38095-38096 for the dimension of R/fR; the conclusion is used inside the counted span.

Source reference. Original proposition 00KQ: sources/stacks-chow/tags.txt, source lines 760–760. Cited at proof lines 38097-38099 for the system of parameters of R/fR; the conclusion is used inside the counted span.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension $d > 1$, let $f \in \mathfrak{m}$ be an element not contained in any minimal prime ideal of R, and let $k \in \mathbf{N}$. Then there exist elements $g_1, \ldots, g_{d-1} \in \mathfrak{m}^k$ such that $f, g_1, \ldots, g_{d-1}$ is a system of parameters.*

Proof. We have $\dim(R/fR) = d - 1$ by Algebra, Lemma 00KW. Choose a system of parameters $\overline{g}_1, \ldots, \overline{g}_{d-1}$ in R/fR (Algebra, Proposition 00KQ) and take lifts $g_1, \ldots, g_{d-1}$ in R. It is straightforward to see that $f, g_1, \ldots, g_{d-1}$ is a system of parameters in R. Then $f, g_1^k, \ldots, g_{d-1}^k$ is also a system of parameters and the proof is complete. □

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